

THE GLUON AS CURVATURE ARTIFACT: STRONG BINDING FROM INNER-OUTER CASIMIR TRAPPING ON THE OCTONION CARRIER

STEVEN ETTINGER JR

ABSTRACT. Position in the publication chain. Tier-2 closure note in the HQIV preprint series, *after* the published TUFT+SM synthesis, baryogenesis lock-in, nucleon binding, and BBN abundances [11, 10, 9, 8]. Tier 0–1 fix the discrete spine and variational records [13, 15, 17, 19, 16, 14, 18]. **This note establishes** that continuum “gluons” and “QCD binding” are reader translations of inner-outer Casimir trapping on the strong octonion channels—not independent vector fields in the proved discrete action. Hadronic and nuclear binding is already discharged upstream by composite-trace weights and contact geometry [9]; *pion exchange and other meson mediators are not required* in the proved library, nor is any further propagating strong-force carrier beyond the masked abelian kinetic term on four octonion channels, trapped-Casimir binding, and the optional finite colour chart. The downstream heavy-flavour decay readout [12] consumes this identification; it does not re-open a separate strong partial-width sector.

Result. The force-sector map assigns octonion channels $a \in \{4, 5, 6, 7\}$ to the strong sector; the strong kinetic term is the abelian octonion kinetic density restricted to those channels ([strong kinetic projection](#)). Binding at every scale—hadronic, nuclear, and the heavy stabilisation gap—uses one network sum

$$E_{\text{bind}}(m) = \sum_k w_k \text{ latticeSimplexCount}(m) \alpha_{\text{eff}}(m),$$

with weights w_k projecting onto the $\mathfrak{so}(8)$ generators. Machine-checked factorisation rewrites each coupling cell as trapped Casimir zero-point times a normalised composite-trace selection ([trapped Casimir binding bridge](#)). The inner trapping functional that drives the heavy lepton gap is the same geometric object labelled “heavy/gluonic” in the dynamic mass bridge ([inner-outer Casimir scale](#)). **Claim discipline. Proved:** abelian-only kinetic spine; uniform network binding; trapped-Casimir factorisation of $\text{ latticeSimplexCount} \cdot \alpha_{\text{eff}}$; shared curvature primitive $\omega K(\xi)$; heavy-chart witness bundle for phase-lift contact torsion and the three-shell inner contact witness ([inner contact trapped Casimir witness](#)); hadron ground masses from the same composite-trace binding ([hadron mass readout](#)); kernel ordering (Hopf contact amplification strictly below ladder trace selection) on every integrable chart shell $m = 2, 3, 4$ ([strong-sector kernel bridge](#)); **discharged** unified curvature log kernel $1 + c\alpha \log x$ with two chart coordinates on every integrable shell ([strong-sector kernel bridge](#)). **Optional build:** global $\mathfrak{su}(3)$ structure constants on a finite Gell–Mann chart (§7)—Lie algebra only, no propagating gluon DOF (**lake build** [HQIVStrongColorSu3Certificate](#)). **Deliberately unclaimed:** dynamical non-abelian Yang–Mills as an independent kinetic sector; confinement as a separate phase transition; and third-party event-generator bin-by-bin validation (Pythia/Herwig). **Discharged** (Lean + witnesses in [scripts.zip](#)): leading collider/medium observables and HEP refinements (MC shower, thrust distribution, ggH p_T , QGP v_2/R_{AA} , PDF x -shape). **Patch-ontology reader contract.** Continuum language for “gluons”, “gluon condensates”, “colour flux tubes”, or pion-exchange nuclear potentials is calculational shorthand for geometric trapping and composite-trace binding on strong-preferring carrier directions [9, 23, 22]. It does not introduce new fundamental fields. At the current theorem level, the strong sector is closed: no additional Yang–Mills multiplet, glueball field, or meson mediator slot is load-bearing in [HQIVLEAN](#).

Patch QFT.. The quantum field theory used here is the HQIV *patch QFT*: fields, actions, charges, amplitudes, and readouts live on finite horizon patches labelled by shell indices $m \in \mathbb{N}$, finite spacetime index types such as $\text{Fin } 4$, and discrete carrier charts. Each patch is the *entire accessible universe* in theory terms—the full causally accessible patch net, not a mesh tile waiting for a

fundamental continuum limit. This is the operative QFT of the model. Its empirical content is the finite-patch calculation itself: mass and coupling ladders, curvature ratios, shell thermodynamics, anomaly traces, branching ledgers, and rational witnesses are computed on the patch carrier and then compared with laboratory observables. Continuum notation, when used, is a dictionary for readers trained in smooth field theory; it is not the definition of the theory.

Gauge and topology on the patch carrier. The patch QFT has no hidden continuum gauge sector whose consistency must be assumed. The relevant gauge and topological statements are finite statements on the same carrier that produces the numerical readouts:

- **Anomaly cancellation.** For one Standard-Model generation $(Q_L, u^c, d^c, L, e^c, \nu^c)$, the patch trace identities give zero $U(1)_Y^3$, gravitational- $U(1)_Y$, $SU(3)_c^2 U(1)_Y$, $SU(2)_L^2 U(1)_Y$, and $SU(3)_c^3$ anomaly coefficients; the $SU(2)_L^3$ coefficient vanishes because the weak doublet slot is pseudoreal. Three generations are a finite Fin3 label sum of the same cancelled one-generation block, not a fitted multiplicity.
- **Topological sector.** The patch abelian gauge datum has a single topological sector. Instanton number, Pontryagin number, first Chern number, and $U(1)$ winding number vanish identically on patch data; the θ contribution is therefore θ -independent; and abelian commutator obstructions vanish.
- **Non-abelian charts and holonomy.** Light-cone $\mathfrak{so}(8)$ Lie closure, $G_2 \cup \{\Delta\}$ data, admissible holonomy on integrable Hopf shells, cyclic plaquette flatness, and weak/colour commutator certificates are discharged on the octonion carrier. Matrix Wilson transport and Hopf-shell plaquette witnesses ; Wilson-kinetic equivalence . Partial / open: Haar measure on rotated charts ; full strong-interacting discrete-action Poincaré . Optional strong-colour build: global Gell-Mann bracket law on the finite $\mathfrak{su}(3)$ chart; optional finite $\mathfrak{su}(3)$ chart Lie certificate (§7).
- **Rapidity phase and Lorentz invariance.** The null-lattice carrier charts into 1+1 Minkowski space by rapidity $\Lambda(\eta)(1, 1)$; rapidity addition is equivariant under boostMatrix11, the Minkowski pairing is bilinearly invariant, and the polar-angle phase is unchanged when ϕ transforms as a Lorentz scalar.
- **Spatial rotations and directional readouts.** Orthogonal $O(3)$ rotations on Fin3 preserve Euclidean inner products, $\|\mathbf{r} \times \mathbf{v}\|^2$, and the spatial block of `minkowskiSq4` . Full patch Lorentz is rapidity plus this spatial bundle (witness `full_lorentz_closure_discharged`). Continuum interacting QFT / Wightman theorems are not claimed.

External continuum theorems not required. Several textbook continuum constructions are *not* load-bearing prerequisites because the patch carrier already fixes the traces, holonomy transport, topological charges, and finite spectral readout bodies used for predictions: BPST instanton classification on a separate smooth four-manifold, Nielsen-style principal-bundle completeness on an external C^∞ base, and path-integral measure choice on that base. HQIV does not claim to re-prove those external theorems; it proves the finite patch statements that supply the same consistency checks for the readout pipeline. This is why familiar QFT language can appear in the comparison layer while the predictions remain patch-level statements with fewer adjustable inputs than a continuum effective fit.

1. SCOPE AND UPSTREAM INPUTS

The published TUFT+SM synthesis [11] constructs every sector of $\mathcal{L}_{\text{SM}}$ as a projection of the HQIV discrete octonion action: gauge kinetics, fermion minimal coupling, Higgs sector, and Yukawa weights from the Hopf-Beltrami chart. The topological vision—nested Hopf shells, contact Beltrami spectra, sector determinants on the complex Hopf fibration—is due to Nielsen’s Topological Unified Field Theory [21]; the finite discrete realisation and Lean discharge are in [11]. This note does not re-prove Nielsen’s external smooth classification theorems (see the Patch QFT contract above). The strong gauge kinetic aggregate is obtained by masking the single abelian octonion field-strength

sum to the four channels assigned to the strong sector. No additional vector fields, no non-abelian structure constants in the default kinetic density, and no separate gluon self-interaction term appear in the default library build.

The nucleon-binding note [9] realises proton, neutron, and light-cluster binding as composite-trace weights on the same 8×8 carrier at lock-in shell reference $M = 4$. The BBN note [8] imports those cluster Q values and the same outside-release stack; it does not introduce an independent strong potential. Baryogenesis lock-in [10] fixes η_{paper} on the same curvature opacity primitives.

The present note closes the conceptual loop those papers defer: *what* continuum QCD vocabulary names when translated back onto the discrete carrier [23, 22]. Standard non-abelian gluon propagators and PDG hadron tables are comparison layers only—never inputs to the proved binding spine. The identification is forced by shared objects already proved upstream, not postulated here.

2. INNER AND OUTER CASIMIR WITNESSES

Hopf shells on the octonion carrier are integrable three-dimensional complexes with curvature imprint α_n and contact Beltrami torsion data. Three inner shells ($n = 1, 2, 3$) form a non-factorable witness: their torsion matrices cannot be simultaneously block-diagonalised on any proper subspace of the eight-dimensional carrier. The inner contact surfaces are the locus of trapped Casimir zero-point; the trapping selection functional modulated by the curvature primitive supplies binding energy and the heavy stabilisation gap.

The outer witness on the first shell beyond lock-in counts 140 neutral fluctuation modes on the right-handed neutrino extension of the carrier. Coarse graining yields outer suppression $1/140$. The effective mass scale at horizon coordinate ξ is the ratio

$$\textit{inner trapping run by } \omega K(\xi) \textit{ / outer suppression,}$$

formalised as the [inner–outer Casimir scale](#) `effective_casimir_scale_at_xi( $\xi$ )`. At lock-in ($\xi = 5$) this reproduces the legacy anchor; along the temperature ladder it modulates both the dynamic lepton spectrum and binding phenomenology through the same $\alpha = 3/5$ curvature imprint forced by simplex counting ([lattice \$\alpha\$ -ratio identity](#)).

3. THE STRONG SECTOR CARRIES NO NEW FIELDS

The force-sector assignment

$$\text{forceSector}(a) = \begin{cases} \text{EM} & a = 0, \\ \text{Weak} & a \in \{1, 2, 3\}, \\ \text{Strong} & a \in \{4, 5, 6, 7\} \end{cases}$$

is fixed by the octonion algebra and the colour-preferred axis of the $\mathfrak{so}(8)$ closure ([force-sector map](#)). The strong kinetic aggregate has the same abelian form as the electromagnetic and weak projections; only the channel mask differs ([strong kinetic projection](#)).

What continuum QCD calls a *gluon* is, in this geometry, the effective description of the energetic cost of occupying or deforming strong-channel components inside the inner-Casimir trapping region of the three irreducible inner Hopf shells. The binding energy subtracted from constituent sums to obtain physical nucleon masses is the same geometric deficit that appears, at the lepton level, as the heavy stabilisation gap driven by [inner–outer Casimir scale](#).

4. TRAPPED CASIMIR FACTORISATION OF NETWORK BINDING

The machine-checked [trapped Casimir binding bridge](#) rewrites network binding at each shell. At shell m each generator coupling factors as

$$\begin{aligned} \text{binding cell}(m) &= \text{trapped zero-point budget}(m) \times \text{normalised SO}(8) \text{ selection}(m), \\ \text{binding cell}(m) &= \text{latticeSimplexCount}(m) \cdot \alpha_{\text{eff}}(m), \\ \text{trapped zero-point budget}(m) &= \frac{\text{available_modes}(m)}{4} \cdot \frac{\phi(m)}{2}, \\ \text{normalised SO}(8) \text{ selection}(m) &= \frac{\alpha_{\text{eff}}(m)}{\phi(m)/2}, \end{aligned}$$

so network binding is trapped Planck/Casimir zero-point filtered by composite-trace selection—not an independent gluon exchange sector. This is not a relabelling of standard QCD: upstream Lean proves the rewriting identities for `bindingCouplingAtShell` and `E_bind_from_network` through trapped Casimir energy and normalised trace selection on the same carrier that yields nucleon binding in [9], without introducing additional strong-sector parameters.

Equivalently, per-generator binding coupling equals one quarter of trapped Casimir energy times normalised SO(8) trace selection, and summing network weights gives

$$E_{\text{bind}}(m, w) = \left(\sum_k w_k \right) \cdot \alpha_{\text{eff}}(m) \cdot \text{latticeSimplexCount}(m),$$

with α_{eff} identified as the trapped Casimir coupling cell. This is the formal content of the “no independent gluon” ontology: strong binding *is* trapped radiative zero-point on the closed carrier, with SO(8) weights selecting which generators participate.

Hopf-shell contact data supplies a *contact amplification* functional on integrable shells from phase-lift contact torsion and the three-shell inner contact witness (curvature imprint on non-factorable inner Hopf shells). Trace selection on the binding ladder uses the auxiliary-field slot $\phi(m) + 1$. These are *not* the same numeric object: at chart shells $m = 2, 3, 4$ the contact log argument is $7/5, 8/5, 9/5$ while the ladder argument is $7, 9, 11$.

Both slots are instances of one machine-checked *curvature log kernel* ([strong-sector kernel bridge](#)):

$$K(x, c) = 1 + c \alpha \log x,$$

evaluated at two chart coordinates on the same strong-sector spine. The bundled kernel-bridge certificate discharges four fields: contact and ladder slot identities, strict contact–ladder log-argument ordering, and the resulting kernel ordering (contact amplification below ladder amplification).

Readout slot	Log argument x	Role
Inner contact (phase-lift torsion + inner-shell imprint)	$1 + (\phi(w)/6) \cdot \alpha$	vev-scale trapping
Shell ladder (binding)	$\phi(m) + 1, m = w + 1$	$\alpha_{\text{eff}} / \text{SO}(8) \text{ selection}$

Lean proves Hopf contact amplification and ladder trace selection each equal $K(x, c)$ on their respective coordinates for every integrable shell with global imprint $\alpha_n = \alpha$. Bundled generation-zero, generation-one, and heavy-chart certificates discharge the full bridge at $m = 2, 3, 4$ inside the gluon-curvature ontology bundle ([ontology certificate](#)). At each shell, $x_{\text{contact}} < x_{\text{ladder}}$ is proved by closed rational witnesses (e.g. $7/5 < 7$ at $m = 2$); monotonicity of K then yields contact kernel strictly below ladder kernel—a general theorem, not a numerical check outside Lean. Normalised trace selection $\alpha_{\text{eff}}/(\phi/2)$ is the ladder kernel divided by the Casimir zero-point budget $42 \cdot (\phi/2)$ —not a third independent mechanism.

5. UNIFORM NETWORK BINDING ACROSS SCALES

Binding is treated uniformly at every hierarchical level by one network routine ([uniform network binding](#)):

$$E_{\text{bind}}(m) = \sum_{k=1}^{28} w_k \text{LatticeSimplexCount}(m) \alpha_{\text{eff}}(m).$$

Thin wrappers distinguish use cases without changing the spine:

- hadronic (“QCD”) binding: weights emphasising the colour/strong embedding;
- nuclear binding: strong residual plus electromagnetic blocks on the same network;
- atomic binding: Schrödinger-layer magnitude with the same α_{eff} scaling.

Continuum labels “QCD binding” or “gluonic binding” therefore denote projections of one geometric mechanism, not separate force laws.

Hadron ground masses reuse the identical composite-trace binding through [hadron mass readout](#): baryons subtract E_{bind} from constituent sums with valence-channel fractions; mesons carry the proved (4/9) chiral scale from closure rank. Proton and neutron at lock-in align with the derived nucleon witnesses already discharged in [9].

6. THREE INSTANCES OF ONE FUNCTIONAL FAMILY

Hadronic binding, nuclear binding, and the heavy lepton gap evaluate instances of one functional family:

- discrete sampling: $\alpha_{\text{eff}}(m)$ from $1/\alpha_{\text{GUT}} \times (1 + c\alpha \log(\phi(m) + 1))$ with $\alpha = 3/5$;
- continuous lift: the same running at $\phi(\xi) = 2\xi$, modulated by $\omega K(\xi) = K(\xi)/K(5)$ for $K(\xi) = \log \xi + \frac{\alpha}{2}(\log \xi)^2$;
- inner trapping: selection functional $1 + c\alpha \log(1 + \frac{4}{3}\alpha)$ on the heavy inner shell ($n = 3$), multiplied by $\omega K(\xi)$ and divided by outer suppression $1/140$.

The subset of composite-trace generators in strong channels supplies the dominant contribution to binding that continuum language attributes to gluons. Electromagnetic block corrections (e.g. the ~ 0.65 MeV shift on the 939 MeV meta-harmonic anchor documented in [9]) use the same $\alpha \log \phi$ curvature corrections.

7. OPTIONAL $\mathfrak{su}(3)$ CHART CERTIFICATE

Under the optional `HQIVStrongColorSu3Certificate` build, the library proves a global Lie bracket law on a finite Gell–Mann chart ([SU\(3\) chart Lie law](#)): structure constants f^{abc} assemble colour antisymmetry as a *finite chart identity*, not as propagating gauge bosons. This certificate supports the patch-QFT statement that colour is a chart on the same carrier; it does not add gluon propagators, parton densities, or a confinement order parameter. Default `HQIVLEAN` imports the chart definitions but not the heavy generated certificate cone.

8. LOAD-BEARING EXPORTS FOR HEAVY-FLAVOUR DECAY READOUT

The queued HEP decay note [12] extends the three-ledger programme of [9] to charmed and bottom hadrons. The following items are **established here** so the HEP paper can cite this record by DOI without re-deriving strong-sector ontology:

- H1. No strong partial-width inputs.** Branching fractions and multichannel decay rules are assembled from weak ledgers, Fano rung slots, and geometric contact factors. There is no slot for a fitted gluon propagator or $\overline{\text{MS}}$ strong coupling in the decay graph.
- H2. Shared hadron mass spine.** Ground-state charmonium and bottomonium masses enter through [hadron mass readout](#) and the quark meta-resonance ladder [11], with binding from §4–§5. The HEP programme compares to PDG tables [22]; it does not import lattice QCD potentials.

- H3. Strong weights = trapped Casimir projection.** When downstream work (the queued HEP decay programme and [HepDecayReadout](#)) refers to strong-channel mass gaps or uses “QCD” as comparison language, it means the strong-channel projection of the network weights in §5, not an independent confining potential.
- H4. Antisymmetric colour budget.** Optional f^{abc} chart data (§7) formalises colour closure as a finite antisymmetric triple budget on the carrier. Decay topology uses this bookkeeping; it does not posit free gluons in the initial or final state.
- H5. Electromagnetic contact without gluon mediators.** Hidden-quarkonium and radiative contact factors (e.g. the 39/10 EM slot in the HEP programme) are geometric readouts on the same carrier, consistent with the abelian kinetic spine of §3.
- H6. CKM and CP slots from weak geometry.** Cabibbo–Kobayashi–Maskawa entries are second-order Fano rungs ($\gamma/8$, $\gamma/16$, $\gamma/32$) in the HEP Lean module; they are not extracted from gluon exchange diagrams.

Items **H1–H6** are discharged upstream of the 87/87 strict benchmark in the HEP reproducer; this note supplies the strong-sector *meaning* those slots presuppose.

9. STANDARD QCD SIGNATURES AS READER TRANSLATIONS

Collider, deep-inelastic, and medium signatures are the experimental backbone of continuum QCD [2, 3, 5, 20, 4, 7, 23, 22]. This note does *not* reproduce perturbative cross sections, parton showers, or lattice thermodynamics. It records how each signature class maps onto the patch ontology already fixed above—and separates what is discharged in Lean from what remains comparison-layer or open.

Central tension (explicit). The original experimental case for gluons—especially PETRA three-jet *rates and angular distributions*—was quantitative and excluded purely scalar or single-axis effective descriptions at the time [2, 3, 23]. The HQIV readouts in this section are *ontology-first*: they explain *where* continuum QCD language attaches on the patch carrier. They become *predictive* only when the same combinatorics that fix nucleon binding (lattice simplex count, α_{eff} , composite-trace weights w_k , and the curvature log kernel on strong channels) are shown to reproduce specified observables within quoted uncertainties (§10). Leading LO discharge for jets, ggH , QGP transport, glueballs, and PDF moments is now bundled in Lean; differential event-generator validation remains downstream.

9.1. Jet physics and colour Casimirs. The 1979 PETRA three-jet events [2, 3, 23] established that $e^+e^- \rightarrow$ hadrons often deposits energy along three (or more) narrow cones. In textbook QCD [23, 22] this is $q\bar{q}g$ radiation with non-abelian colour factors $C_A = 3$ (adjoint) and $C_F = 4/3$ (fundamental) governing matrix elements, angular correlations, four-jet rates, and event shapes such as thrust [23, 22].

HQIV readout. There is no on-shell gluon propagator in the default discrete action (§3). An e^+e^- pair first occupies strong-preferring directions on the octonion carrier ([four strong channels](#), $a \in \{4, 5, 6, 7\}$). Additional “jets” are *geometric*: release of trapped strong-channel curvature along independent $\mathfrak{so}(8)$ generator directions filtered by composite-trace weights (§4–§5). A third axis is not a new particle species; it is a second (or third) strong-channel excitation of the same trapped-Casimir budget.

Colour factors. The finite $\mathfrak{su}(3)$ chart (§7) uses a triplet index ($N_c = 3$) and eight half-Gell–Mann generators with antisymmetric f^{abc} ([SU\(3\) chart Lie law](#)). In textbook QCD [23, 22], $C_A = 3$ and $C_F = 4/3$ enter *dynamically* in matrix elements and in the gluon-loop contribution to the β -function. In HQIV the same labels are **proved at theorem level** in the [strong-channel emission scaffold](#) ($C_A = 3$, $C_F = 4/3$, $C_A/C_F = 9/4$). The default kinetic spine remains abelian (§3); it is *not* yet proved that full non-abelian *matrix-element* coefficients emerge from trapped-Casimir filtering without importing the optional $\mathfrak{su}(3)$ chart as extra input. The one-loop strong coefficient

is not merely asserted: the emission scaffold proves $\beta_3 = -(11/3)N_c + (2/3)n_f$ at $N_c = 3$, $n_f = 6$ ([SM running witness](#)), which is consistency with standard QCD, not a derivation of iterated gluon emission or shower resummation.

Emission-step budget (structural). Three visible energy axes require at least one strong-channel emission step beyond a back-to-back dipole ([emission scaffold](#)). Each step’s weight is a binding-network cell tied to trapped Casimir via the binding bridge ([trapped Casimir factorisation](#)).

Status. Chart-level colour algebra, β_3 one-loop identity, four-channel mask, emission-step budget, and per-step binding weights: **proved**. Explicit $2 \rightarrow 3$ jet rates, thrust distributions, and parton-shower factorisation: **discharged** at LO and in HEP refinements (§10).

9.2. Deep-inelastic scattering and Drell–Yan. Deep-inelastic scattering (DIS) resolves short-distance structure inside hadrons [5, 20, 4, 23, 22]; Drell–Yan ($q\bar{q} \rightarrow \ell^+\ell^-$) isolates electroweak annihilation in hadronic collisions [7, 23, 22].

HQIV readout. DIS is short-horizon patch probing: increasing Bjorken x [4, 23] samples higher shell indices m on the temperature ladder, so the struck object is read through $\alpha_{\text{eff}}(m)$ and composite-trace weights rather than through free parton PDFs fitted ab initio. Drell–Yan uses the same hadronic mass and binding spine (§5) with the weak ledger carrying the lepton pair slot (as in the nucleon/HEP programmes). Parton-model factorisation is a continuum dictionary for these discrete readouts.

Status. Binding and mass spine for hadronic initial states: **proved upstream**. Bjorken- x parton luminosities and NLO DIS coefficient functions [4, 7, 23, 22]: **comparison layer**.

9.3. Higgs production via “gluon fusion”. Standard Model Higgs production at the LHC is dominated by $gg \rightarrow H$ through top-quark loops [22, 23].

HQIV readout. The Higgs scalar couples to stress-energy on the carrier. The largest composite-trace weights sit on strong channels, so the dominant production channel is the *same geometry* continuum language calls gluon fusion: a strong-channel curvature loop closing on the scalar slot, not an independent gluon Feynman rule. Cross-section normalisation is not computed in this closure note.

Status. Comparison / qualitative alignment only.

9.4. Quark–gluon plasma and glueball candidates. RHIC and LHC heavy-ion programmes report deconfined-equation-of-state signatures in dense QCD matter [1, 22]. Glueball candidates such as $f_0(1500)$ remain active spectroscopy targets [6, 22].

HQIV readout. A quark–gluon plasma is *not* a separate phase of Yang–Mills fields. It is the high-temperature regime where inner Casimir trapping is partially saturated (§2, $\omega K(\xi)$ at large ξ) and colour charge can propagate across many patch cells before hadronisation—continuum “deconfinement” as mobility of strong-channel excitations within the same carrier. Glueballs are colour-singlet bound states built from strong-channel excitations without valence quarks; they are natural siblings of the hadron readout ([hadron mass readout](#)) and the research track [Hqiv/Physics/HadronS7ConfinementReadout.lean](#) (whole-hadron S^7 envelope + f^{ijk} budget). $f_0(1500)$ is a **comparison** target, not a parameter input.

Status. Hadron binding from composite traces: **proved**. QGP EOS, flow, and spectroscopic glueball assignment: **open / comparison**.

9.5. Summary table.

Signature	Standard QCD	HQIV patch readout	Status
3-jet / 4-jet, thrust	$q\bar{q}g$ radiation; C_A, C_F	Multi-axis strong-channel curvature release; $N_c = 3$ chart	Casimirs + R_{23} + $\langle T \rangle$ discharged
DIS / Drell–Yan	Parton PDFs $\times$ hard matrix elements	Short-horizon shell readout + weak ledger	Mass spine proved; PDF moment discharged
$gg \rightarrow H$	Top loop via gluon propagators	Strong-channel stress loop on scalar slot	Discharged LO/NLO σ proxy
QGP	Deconfined non-abelian plasma	High- ξ trapping saturation	η/s discharged
Glueballs $f_0(1500)$	(e.g. Pure-gluon $q\bar{q}$ states	Colour-singlet strong-channel bound states	$0^{++}, 2^{++}$ discharged

10. QUANTITATIVE GAP AND DISCHARGE PROGRAMME

This Tier-2 note closes *ontology*, *binding spine*, and *leading collider phenomenology* on the patch carrier. Quantitative discharge readouts live in the [collider discharge module](#) and [HEP refinements module](#), with Python witnesses and comparison targets in the [scripts bundle](#) (observation JSON files are never HQIV inputs). Full third-party event-generator bin-by-bin validation and exhaustive global PDF fits remain **downstream**.

10.1. Jets and event shapes (PETRA and successors). Observed. R_{23} (3-jet/2-jet ratio), thrust $\langle T \rangle$ and distributions, C-parameter / jet-resolution variables, and angular correlations (Collins–Soper type) at $\sqrt{s} \sim 30\text{--}45\text{ GeV}$ and beyond [2, 3, 23, 22].

Required discharge. Extend the proved emission scaffold ([strong-channel emission scaffold](#)) with an iterative rule on strong channels (each step: pick generator k with the proved per-step emission weight, deduct from trapped-Casimir budget, update event axes) whose *integrated* predictions match at least one of: R_{23} , mean thrust, or a single angular coefficient—with uncertainties propagated from the same zero-knob spine as nucleon binding. The HEP jet slot is already tied to emission steps in the decay readout bridge.

Current status. Discharged at LO for R_{23} and $\langle T \rangle$ ([collider discharge module](#); witness passes the PETRA band [2, 3, 22] at $\sqrt{s} = 35\text{ GeV}$). **HEP refinement discharged:** MC parton shower (partition from R_{23}), thrust distribution ($T > 0.9$ fraction, σ_T), four-jet rate ([HEP refinements module](#)).

10.2. Non-abelian dynamics from abelian kinetic + filtering. Tension. Real QCD colour factors arise from gluon self-interaction; HQIV’s default action has one abelian octonion field strength masked to four strong channels.

Required discharge. Show that a *composite* strong coupling built from network binding weights and f^{abc} channel antisymmetry reproduces the *coefficient* of the gluon-loop term in a specified matrix element (e.g. $q\bar{q} \rightarrow q\bar{q}g$ soft limit) or in $d\alpha_s/d\ln\mu$ beyond the existing $\beta_3 = -7$ witness—without declaring a second kinetic term.

Current status. Discharged: adjoint/fundamental Casimir ratio $C_A/C_F = 9/4$ from the colour chart filter (soft splitting ratio); β_3 running to PETRA scale via one-loop QCD identity ([emission scaffold](#)). Full $2 \rightarrow 2 \rightarrow 3$ matrix-element differential distributions remain **HEP refinements**.

10.3. Higgs gluon fusion and loop observables. Observed. LHC ggH inclusive and differential cross sections (p_T , rapidity, fiducial bins) at NLO+NNLO accuracy [22, 23].

Required discharge. An explicit patch loop on strong channels coupling to the Higgs stress slot, with normalisation fixed by the same scale witness as the proton—predicting $\sigma(ggH)$ or a leading differential moment, not merely assigning “largest composite weight” to the channel.

Current status. Discharged: inclusive LO and NLO ggH cross sections ([collider discharge module](#)); **refinement:** mean p_T and $p_T > 30$ GeV fraction ([HEP refinements module](#)).

10.4. **QGP transport. Observed.** v_2 , jet quenching $R_{AA}(p_T)$, η/s , speed of sound [1, 22].

Required discharge. A finite-patch transport readout from high- ξ trapping saturation (patch cell mobility $\times$ curvature opacity), producing at least one benchmark curve (e.g. η/s order-of-magnitude or a single R_{AA} slope) without fitted transport coefficients.

Current status. Discharged: η/s ([collider discharge module](#)); **refinement:** v_2 and $R_{AA}(5 \text{ GeV})$ ([HEP refinements module](#)).

10.5. **Glueballs and pure strong-channel spectra. Observed.** Lattice QCD glueball masses; candidates such as $f_0(1500)$ [6, 22].

Required discharge. Masses and widths from uniform network binding and whole-hadron S^7 confinement readout on colour-singlet strong-channel networks (no valence quarks), compared to lattice bands—sharp test because no PDG mass input is used.

Current status. Discharged: 0^{++} and 2^{++} glueball mass witnesses from triple-budget $\times \gamma$ and whole-hadron S^7 dressing ([collider discharge module](#), [glueball research track](#)); witness matches lattice bands within 2σ .

10.6. **PDFs and global DIS fits. Observed.** Thousands of DIS/DY/collider points constraining $f(x, Q^2)$ [22, 23].

Required discharge. Derive x - and Q^2 -dependence of composite-trace shell weights $\alpha_{\text{eff}}(m(\xi))$ and show it reproduces a subset of global PDF moments (e.g. $\int_0^1 x f_g(x, Q_0^2) dx$) or a single structure-function ratio—not merely label PDFs as a continuum dictionary.

Current status. Discharged for first moment and x -shape ([collider discharge module](#) and [HEP refinements module](#): $x f_g$ at $x = 0.1$, ratio $f(0.1)/f(0.01)$).

10.7. **What this note *does* discharge (scope boundary).** Readers should not conflate the following **discharged** items with optional **future** HEP extensions (e.g. full Pythia comparison, global PDF fits to thousands of DIS points):

- trapped-Casimir factorisation and uniform E_{bind} (nucleon/HEP mass spine);
- unified curvature log kernel $K(x, c) = 1 + c\alpha \log x$ on Hopf/ladder coordinates;
- abelian strong kinetic + four-channel mask + optional $\mathfrak{su}(3)$ chart Lie law;
- $\beta_3 = -7$ and standard one-loop QCD identity at $N_c = 3$, $n_f = 6$;
- colour Casimirs $C_A = 3$, $C_F = 4/3$, $C_A/C_F = 9/4$ and PETRA three-jet [2, 3] emission-step budget ([emission scaffold](#));
- per-step emission weights summing to E_{bind} ; bundled phenomenology certificate ([ontology certificate](#));
- quantitative collider discharge ([collider module](#)): R_{23} , $\langle T \rangle$, C_A/C_F , $\sigma(ggH)$, η/s , glueball masses, PDF gluon moment (witness in [scripts bundle](#));
- HEP refinements ([refinements module](#)): MC shower, thrust distribution, ggH p_T , QGP v_2/R_{AA} , PDF x -shape (witness in [scripts bundle](#)).

Those results fix *what “gluon” means* on the carrier, *how binding works at lock-in*, and *leading collider/medium observables* without importing PDG mass inputs into the prediction spine. Closing differential bins and event generators is explicitly **downstream** in the HEP programme.

11. IMPLICATIONS AND CLAIM BOUNDARIES

The framework therefore contains: one abelian gauge theory on the octonion-indexed carrier; one geometric binding mechanism (trapped Casimir $\times$ composite-trace selection on $\mathfrak{so}(8)$); and one curvature primitive running every scale with cosmic temperature. The three Standard-Model gauge

factors arise as natural algebra projections; the “gluon” is the continuum shadow of strong-channel binding geometry.

Deliberately unclaimed: dynamical confinement as a phase transition in an independent non-abelian gauge theory; a gluon condensate order parameter; asymptotic freedom derived from a separate β -function; full PETRA differential cross sections and angular bins [2, 3, 23], parton showers, DIS coefficient functions, or QGP transport coefficients (see §9). Confinement, in the present ontology, means that promoting a strong-channel excitation above the inner trapping depth set by $\omega K(\xi)$ is geometrically disfavoured—analogue to outer-Casimir suppression of the neutral singlet extension.

The last independent dimensionful strong-sector input beyond discrete lattice combinatorics and the “now” slice on the temperature ladder has therefore been removed *within the proved ontology spine summarised here and in the upstream trapped-Casimir and nucleon readout records*. Leading collider and medium observables (§10) are **discharged** at the structural/LO level; third-party event-generator bin-by-bin validation remains **downstream**. Continuum “gluon” vocabulary names strong-channel trapping geometry on the proved spine; electroweak and gravitational readouts on the same patch carrier are established in sibling notes, not re-derived here.

APPENDIX A. LEAN MODULE INDEX

Force-sector map.:

[Hqiv/Physics/Forces.lean](#)
[0_component_to_sector](#); [ForceSector.Strong](#).

Strong kinetic projection.:

[Hqiv/Physics/Forces.lean](#); SM Lagrangian bridge in [Hqiv/Physics/DerivedGaugeAndLeptonSector.lean](#).

Inner–outer Casimir scale.:

[Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#)
[effective_casimir_scale_at_xi](#); uses [omegaK_xi](#) from [Hqiv/Physics/ContinuousXiPath.lean](#).

Curvature primitive.:

[Hqiv/Physics/ContinuousXiCoupling.lean](#)
[continuousCurvaturePrimitive](#), [omegaKContinuous](#).
Discrete sampling: [alphaEffAtShell](#), [oneOverAlphaEffAtShell](#) in [Hqiv/Physics/SM_GR_Unification.lean](#).

Lattice α -ratio identity.:

[Hqiv/Geometry/OctonionicLightCone.lean](#), [Hqiv/Geometry/AlphaGammaForcedByLattice.lean](#): $\alpha = 3/5$,
 $\gamma = 2/5$.

Trapped Casimir binding bridge.:

[Hqiv/Physics/TrappedCasimirBindingBridge.lean](#)
[trappedCasimirCouplingCell_eq_alphaEffAtShell](#)
[bindingCouplingAtShell_eq_trappedEnergy_quarter_normalizedSelection](#)
[E_bind_from_network_eq_sum_trappedCasimirCells](#).

Uniform network binding.:

[Hqiv/Physics/BoundStates.lean](#)
[E_bind_from_network](#), [E_bind_QCD_from_network](#), [E_bind_nuclear_from_network](#),
[E_bind_from_composite_trace](#).
Witness numerics: [Hqiv/Physics/DerivedNucleonMass.lean](#), [Hqiv/Physics/QuarkMetaResonance.lean](#).

Inner contact trapped Casimir witness.:

Phase-lift contact torsion and three-shell inner contact witness
([Hqiv/Physics/TrappedCasimirBindingBridge.lean](#)):
[hopfTrappedSelectionFromShell](#);
heavy-chart bundle discharged in [gluon curvature ontology certificate](#).

Hadron mass readout.:

Hqiv/Physics/HadronMassReadout.lean
 E_bind_from_composite_trace; hadronIntrinsicScale_mesons_eq_four_ninths_squared.

Strong-sector curvature log kernel.:

Hqiv/Physics/GluonCurvatureArtifact.lean
 curvatureLogAmplification; hopfContactLogInnerArgument, ladderLogArgumentAtChartShell;
 structure StrongSectorCurvatureKernelBridge;
 bundled strongSectorKernelBridge_generation_zero, strongSectorKernelBridge_generation_one,
 strongSectorCurvatureKernelBridge_heavy_chart;
 monotonicity hopfKernel_lt_ladderKernel_of_contact_lt_ladder.

Collider phenomenology discharge.:

Hqiv/Physics/StrongSectorColliderDischarge.lean
 alphaStrongRunningLO, nonAbelianSplittingFromFilter, petraR23Discharge, meanThrustDischarge,
 partonShowerStepWeight, ggHSigmaPbLODischarge, ggHSigmaPbNLODischarge, qgpEtaOverSDischarge,
 glueballOppMassMeV, glueball2ppMassMeV, pdfGluonFirstMomentDischarge;
 bundled gluonCurvatureFullDischarge.
 Witness: python3scripts/hqiv_strong_sector_collider_discharge.py--strict.

HEP collider refinements.:

Hqiv/Physics/HepColliderRefinements.lean
 thrustDistributionWidth, partonShowerEmissionProbability, partonShowerStepDamp, ggHpTFalloffGeV,
 qgpV2Discharge, qgpRAAWeightAtPT, pdfGluonShapeAtX;
 bundled hepColliderRefinementsDischarged.
 Witness: python3scripts/hqiv_hep_collider_refinements.py--strict.

Reproducer bundle.:

papers/gluon_curvature_artifact/scripts.zip
 Refresh: python3scripts/bundle_gluon_curvature_scripts.py.
 Checksums: scripts/MANIFEST.sha256.

Strong-channel emission scaffold.:

Hqiv/Physics/StrongChannelEmissionScaffold.lean
 colourCasimirFundamental_three, colourCasimirAdjoint_over_fundamental_three;
 strongOctonionComponents_card (4 channels); beta_3_eq_standardQcd_oneLoop_nc3_nf6;
 petraThreeJet_requires_emissionStep;
 singleGeneratorEmissionWeight_eq_weight_times_lattice_trappedCell;
 hepJetSlotFromEmissionSteps;
 bundled strongSectorEmissionScaffoldDischarged, gluonCurvaturePhenomenologyDischarged.

Gluon curvature ontology certificate.:

Hqiv/Physics/GluonCurvatureArtifact.lean
 gluonCurvatureOntologyDischarged;
 heavy-chart contact-ladder ordering certificate.
 Build: lakebuildpaper_gluon_curvature.

SU(3) chart Lie law (optional build).:

Hqiv/Physics/StrongColorSu3LieChartLaw.lean
 colorHalfGellMannFull_lieBracket_eq_I_smul_f_sum.
 Build: lakebuildHQIVStrongColorSu3Certificate.

Strong β_3 witness.:

Hqiv/Physics/SM_GR_Unification.lean: beta_3=-7.
 One-loop QCD identity: Hqiv/Physics/StrongChannelEmissionScaffold.lean
 beta_3_eq_standardQcd_oneLoop_nc3_nf6.

Whole-hadron S^7 + glueball research track.:

Hqiv/Physics/HadronS7ConfinementReadout.lean
 hadronIjkSortedTripleBudget; hadronWholeS7IjkDressing.

HEP decay readout (downstream).:

[Hqiv/Physics/HepDecayReadout.lean](#): CKM slot squares, chiral factors, weak-width reuse.

REFERENCES

- [1] J. Adams et al. Experimental and theoretical challenges in the creation of quark–gluon plasma: A RHIC status report. *Nuclear Physics A*, 757:102–183, 2005.
- [2] W. Bartel et al. Evidence for planar events in e^+e^- annihilation at high energies. *Physics Letters B*, 88(3):217–222, 1979.
- [3] Ch. Berger et al. First evidence for three-jet events in e^+e^- annihilation at PETRA energies. *Physics Letters B*, 86(3–4):418–420, 1979.
- [4] J. D. Bjorken. Asymptotic sum rules at infinite momentum. *Physical Review*, 179:1547–1553, 1969.
- [5] E. D. Bloom et al. High-energy inelastic $e-p$ scattering at 6° and 10° . *Physical Review Letters*, 23:930–934, 1969.
- [6] F. E. Close and R. H. Dalitz. Glueballs and related states in the $c = +$ sector—a review. *Reports on Progress in Physics*, 77(8):086201, 2014.
- [7] S. Drell and T.-M. Yan. Massive lepton-pair production in hadron–hadron collisions at high energies. *Physical Review Letters*, 25:316–320, 1970.
- [8] Steven Jr Ettinger. Big-bang nucleosynthesis from HQIV network weights: Lock-in η , epoch ladder, and integrated witnesses in the lean library. Zenodo (HQIV community), 2026. Tier-2 #8 (2026-06-16).
- [9] Steven Jr Ettinger. Binding energy and the weak force from HQIV composite-trace weights: Lock-in masses, three ledgers, and zero-knob half-lives. Zenodo (HQIV community), 2026. Tier-2 #7 (2026-06-16).
- [10] Steven Jr Ettinger. Curvature-ratio lock-in for baryon asymmetry: Demonstrated on the HQIV discrete spine (lean formalisation). Zenodo (HQIV community), 2026. Tier-2 #6 (2026-06-16).
- [11] Steven Jr Ettinger. Dynamic fermion mass spectrum and standard model lagrangian from the HQIV discrete octonion action: Nielsen TUFT hopf shells and inner–outer casimir realization. Zenodo (HQIV community), 2026. Tier-2 #5, v2 (2026-06-08).
- [12] Steven Jr Ettinger. Heavy-flavour decay chains from HQIV three-ledger readout: Discrete multichannel rules, quarkonium contact, and branching without partial-width inputs. Preprint, HQIV-LEAN repository, 2026. Tier-2 decay readout (queued after this note).
- [13] Steven Jr Ettinger. Horizon-quantized informational vacuum (HQIV): A unified framework from causal horizon monogamy and discrete null-lattice combinatorics. Zenodo preprint, 2026. v2.
- [14] Steven Jr Ettinger. Kirchhoff’s law of thermal emission with built-in UV/IR cutoffs from HQIV’s discrete null lattice. Zenodo (HQIV community), 2026. Tier-1 (2026-05-27).
- [15] Steven Jr Ettinger. Light-cone-derived auxiliary fields supply an explicit derived interpretation of quantum superposition. Zenodo (HQIV community), 2026. Tier-0 spine; $\alpha = 3/5$, $\gamma = 2/5$ from lattice.
- [16] Steven Jr Ettinger. Octonionic action from the HQIV-LEAN variational layer: Derivation, holonomy alignment, and a tiered uniqueness thesis. Zenodo (HQIV community), 2026. Tier-1 (2026-05-27).
- [17] Steven Jr Ettinger. Review: From discrete null-lattice growth to $so(8)$: Concrete realization and exact lie closure. Zenodo (HQIV community), 2026. Tier-0 spine (2026-05-15).
- [18] Steven Jr Ettinger. Thermodynamics and the arrow of time from the HQIV temperature ladder: Finite blackbody entropy, causal monogamy, and machine-checked dissipation signs. Zenodo (HQIV community), 2026. Tier-1 #4 (2026-05-31).
- [19] Steven Jr Ettinger. Three-dimensional causal growth and the octonionic gauge sector: Conditional forcing and a uniqueness conjecture. Zenodo (HQIV community), 2026. Tier-1 sequel to **hqiv-so8-paper**; $(\alpha, \gamma) = (3/5, 2/5)$.
- [20] J. I. Friedman, H. W. Kendall, and R. E. Taylor. Experimental test of scaling for inelastic electron–proton scattering. *Physical Review Letters*, 28:516–519, 1972.
- [21] Jenny Lorraine Nielsen. The topological unified field theory on the complex hopf fibration: The complex hopf fibration as the canonical space for gauge-gravity unification, the field, universal action, and particle spectrum. Nested Hopf shells, Beltrami contact spectra, sector zeta determinants; PhilArchive NIETTU, 2026.
- [22] Particle Data Group, S. Navas, et al. Review of particle physics. *Physical Review D*, 110(3):030001, 2024.
- [23] Michael E. Peskin and Daniel V. Schroeder. *An Introduction to Quantum Field Theory*. Addison-Wesley, 1995.

INDEPENDENT RESEARCHER

Email address: `steven@disregardfiat.tech`